Comparative Analysis of Two Rapid Malaria Diagnostic Tests Against the Gold Standard and Implications for Malaria Treatment in Northcentral and Northwestern Nigeria

Ibrahim T,[1*] Musa FM,[2] Umar H[1] Ebong E1[3]

ABSTRACT

Malaria is one of the most prevalent human parasites of public health concern as it can cause mortality especially in children. This study was aimed at undertaking a comparative performance of two popular brands of RDT kits used in Keffi and Kaduna against the gold standard, Microcopy. A total of 655 febrile consenting adults were selected using simple random technique. Blood samples were collected and screened for malaria parasite infection. Malaria parasite tests were conducted using Rapid diagnostic test (RDT) and microscopy technique. The results revealed that in Nasarawa State, Carestat and SD Bioline had similar sensitivity of 45.1%, specificity of 96.6%, and positive predictive value of 89% and negative predictive value of 74%. In Kaduna, Carestat and SD Bioline had sensitivity of 91% & 96% respectively, specificity of 98% & 100%, positive predictive value of 95% & 99% and negative predictive value of 98% & 99% respectively. Thus, the high diagnostic performance of Carestat and SD Bioline indicates that they can be effectively used to diagnose malaria infection. The prevalence of malaria was higher in Kaduna State 167(39.9%) than in Nasarawa State 152(40.3%). Percentage prevalence was higher among females [100(45.5%)] than males [52(33.1%)] in Nasarawa State but for Kaduna prevalence was higher in males 97(65.5%) than females 70(53.8%) ($p \leq 0.05$). However, Microscopy was able to detect the highest number of positive malaria cases in Kaduna 94 (33.8%) and Nasarawa States 79(21% respectively). This thus implies that microscopy remains a better diagnostic tool for malaria parasite detection than the RDTs.

Keywords: *Malaria, Microscopy, RDT, Sensitivity, Kaduna, Nasarawa*

INTRODUCTION

Malaria is a mosquito-borne disease caused by the apicomplexan protozoan from the genus *Plasmodium* that is transmitted by the female mosquitoes of the genus *Anopheles*.[1] Malaria is one of the most prevailing human parasitic infections taking a position of priority in terms of its socioeconomic and public health importance in sub-Saharan African and south east Asia[2].

Department of Science Laboratory Technology,[1] Nasarawa State University, Keffi. Department of Microbiology,[2] Kaduna State University, Kaduna, Nigeria. Department of Medical Laboratory Science,[3] University of Uyo Teaching Hospital, Uyo

***Corresponding Author**: ibrahimtaibat@nsuk.edu.ng
Date manuscript was received: 17/6/23
Date manuscript was accepted: 21/12/23

With regards to control of malaria parasite infection, remarkable and significant gains were achieved in terms of cases and mortality reduction worldwide (from 585,000 to 405,000) between 2010–2018, a consequence of deliberate intervention strategies.[3] According to the WHO, the number of malaria endemic countries dropped from 108 to 91, inclusive of ten certified malaria-free countries and 29 thwarting reappearance of malaria infections during the same period[3]. In Nigeria, malaria parasite infection remains a significant public health burden and accounts for 27% of the total disease burden in Nigeria.[4] Rapid and effective diagnostic approaches assuage suffering, and decreases community transmission.[5] Microscopy is an essential tool for malaria diagnosis and remains the gold standard.[6] However, some of the disadvantages of microscopy in diagnosis include difficulty determining the species of the implicating *Plasmodium* parasite.[7] In addition, microscopy approach is labor intensive and time consuming, has lower detection limit of between 4–20 parasites/ml even for expert microscopists[8] and challenges with electricity supply in parts of Nigeria. Despite the shortcomings highlighted, microscopy remains the gold standard for diagnosis because it has the capacity to identify patients with active malaria and provides information on the parasite density, for monitoring the efficacy of treatment.[9]

Malaria rapid diagnostic tests (RDTs) remain essential tools for case management and surveillance in Nigeria and worldwide. In Nigeria, studies using nationally representative data report sensitivity of about 87–88% and specificity around 75–76% when compared to expert microscopy, with performance varying by age, anaemia status, and parasite density.[10] Globally, most conventional RDTs meet WHO procurement criteria, showing sensitivity between 75–90% and specificity above 93% under field conditions.[11] While HRP2-based tests are highly effective for *Plasmodium falciparum*, their accuracy decreases in low-parasitaemia or non-falciparum infections, and false negatives may occur due to pfhrp2/3 gene deletions, which are currently rare in Nigeria.[12] Recent ultrasensitive RDTs (uRDTs) have shown improved detection of low-density and asymptomatic infections, although molecular tests remain more sensitive.[13] These findings highlight the need for ongoing quality monitoring, confirmatory testing in clinically suspicious cases, and adherence to WHO standards for procurement and implementation.[11]

World Health Organization (WHO) recommends prompt malaria diagnosis either by microscopy or rapid diagnostic tests (RDTs) for all patients with suspected malaria before they are given treatment. Early and accurate diagnosis is essential both for effective management of malaria and other febrile illnesses and for strong malaria surveillance.[14] Rapid Diagnostic Tests (RDTs) are immune-chromatographic-based diagnostic tests. Antigens of the parasite are detected when blood is dropped in a membrane strip, where the liquid sample flows across, bringing the antigens to specific antimalarial antibodies and producing a visible colorimetric change on the nitrocellulose strip. RDTs can be considered fast (Point of care) PoC tests: they are cost-effective, easy to use, and easy to interpret; results are provided in a few minutes; and no electrical equipment is required neither does it require intense training but few personnel.[1-2]

Comparative investigation on malaria diagnosis using RTDs versus microscopy in North central and western Nigeria is lacking in literature. Proper screening and diagnosis of affected persons is an important step in halting disease transmission. This study was therefore aimed at evaluating the Prevalence of Malaria parasite infection among febrile adults attending Nasarawa State University hospital, Keffi and General Hospital Kawo, Kaduna State as well as the comparative analysis of RDT against Microscopy in terms of sensitivity and specificity.

MATERIALS AND METHODS

Study areas

Samples were collected from two Northern States health care centers of Nigeria (Nasarawa and Kaduna). Nasarawa State lies in Latitude eight 5^0 N of the equator and Longitude seven 8^0 E, it also situates on altitude of 850M above sea level.[15] Kaduna state is located between latitude 10-degree centi 3858" N and 10-degree centi 25'36" North of the equator to longitude 7-degree centi grade 22'14" E and 7degree centi grade 32'00" East of the Greenwich Meridian.[16]

Study Population

Samples were collected from febrile consenting adults at the outpatients' department of the Nasarawa State University

Medical Centre, Keffi and General Hospital Kawo, Kaduna State. Ethical approval was obtained from the research and ethics committee of Nasarawa State University, keffi.

Sample Size Determination

The sample size was determine using the formula by Naing & Rusli [17] for sample size calculation at 0.05 level of precision:

$$n = \frac{Z^2pq}{d^2} \quad \text{Equation 1}$$

Where:

n = desired minimum sample size if the target population is > 10,000.

Z= standard normal deviation at the required confidence interval (1.96) which corresponds to 95% confidence interval.

P = Prevalence rate 56.8%[18] for Nasarawa State; 36.7%[19] for Kaduna State).

q=I-p

d= degree of accuracy/precision expected i.e. 0.05 and therefore; p=0.1

$$n = \frac{1.96^2(0.568)(0.432)}{(0.05)^2}$$

$$= \frac{3.8416 \times 0.245}{0.0025}$$

$$= \frac{0.943}{0.0025}$$

= 377

$n = 377$

Thus, a total of 377 samples for Nasarawa State and 278 samples for Kaduna State.

Collection of Sample

A total of 655 blood samples were collected from the patients by venipuncture following informed consent and ethical approval.

Eligibility Criteria

Patients presenting with clinical symptoms of malaria and visiting hospitals in any of the two locations between 7th February and 10th April 2022. Patients with clinical symptoms of malaria determined by a headache, body aches and temperature of 37.5°C and above were included in the study. Individuals who recently took anti-malarial drugs were excluded from the study.

Blood Collection and Laboratory procedures

About 3 ml of blood sample was obtained from each participant using venipuncture method and transferred into an ethylene-diamine-tetra-acetic acid (EDTA) tube to prevent blood coagulation. Next, thick blood smears were made on well cleaned grease free slides and subsequently, stained in the method described by Cheesebrough.[20] The slides were later viewed by two certified medical laboratory scientists with experience in malaria microscopy to ensure validity of results using x100 objective lens of the light microscope to confirm the presence or absence of *Plasmodium* parasites and the species present.

Rapid diagnostic test (RDT)

Blood samples collected from the Patients were also tested for malaria parasite using carestat (Access Bio, U.S.A) and SD Bioline (Abbott, South Korea) carefully following manufacturer's protocol. The kits were stored as instructed by the manufacturer. In this study, the RDTs used were based on HRP-2 and detects only *P. falciparum*.

Determination of sensitivity and specificity

1. Sensitivity (Positive for disease): Sensitivity is the ability of a test to correctly classify an individual as 'diseased'.
   Calculation of sensitivity = TP/TP + FN*100

2. Specificity (Negative for disease): The ability of a test to correctly classify an individual as a disease-free is called the test's specificity.
   Calculation of Specificity = TN/FP + TN*100 = Probability of being test negative when disease is absent.

3. Positive Predictive Value (PPV): It is the percentage of patients with a positive test who actually have the disease. PPV tells us about the number of test positives that are true positives; and if these numbers are higher (as close to 100 as possible), then it suggests that this new test is doing as good as 'gold standard'.
   PPV = TP/TP + FP *100
   PPV = TP (true positive)/TP + FP (true positive + false positive) = Probability (patient having disease when test is positive).

4. Negative Predictive Value (NPV): It is the percentage of patients with a negative test who did not have the disease. NPV tells us how many of test negatives are true negatives; and if this number is higher (should be close to 100), then it suggests that this new test is doing as good as 'gold standard'.
   Calculation of NPV = TN/TN + FN*100
   = TN (true negative)/FN + TN (false negative + true negative) = Probability (patient not having disease when test is negative).
   Where TP = True positive, FP= False positive, FN= False negative, TN =True negative

Statistical Analysis

Data were entered in Microsoft Excel 2013. Distribution of participants' characteristics and malaria parasite prevalence were assessed using contingency tables. Taking blood slide microscopy as the gold standard, the performance of the Rapid Diagnostic Test was compared to it by computing the sensitivity, specificity, the negative predictive value, and the positive predictive values. Data are presented in figures and percentages. Comparison between two means was done using the students'-test analysis.

RESULTS

Prevalence of Malaria in the study population

The Percentage prevalence of malaria parasite infections in blood sample of patients at the two study locations is as presented in Table 1. Overall prevalence of malaria was higher in Kaduna State [167(39.9%)] compared to Nasarawa State [152(40.3%)] with statistical significance at $p \leq 0.05$. However, for gender, the percentage prevalence was higher among females [100(45.5%)] than males [52(33.1%)] in Nasarawa State (p=0.016). Conversely, for Kaduna State, malaria prevalence was higher [97(65.5)] in male subjects than females [70(53.8%] with statistical significance at $p \leq 0.05$.

Malaria Positivity by Microscopy ad RDTs in the study locations

The relative malaria positivity by RDTs kits and microscopy is as shown in Table 2. The result shows that the Microscopy technique had [94(33.8%)] positive results in Kaduna State and [79(21.0%)] positive results in Nasarawa State. The RDTs positive results obtained in both Kaduna and Nasarawa States was 73(19.4%).

Relative malaria percentage positive and negative by RDT kits and microscopy in Nasarawa and Kaduna States are as presented in Table 3. The RDTs used in this study revealed an equal percentage positive result of 73(93.4%) for both Care Stat and SD Bioline. However, malaria positive results using Care Stat was [73(26.3%)] and Bioline [70(25.2%)]. The case was however different for Kaduna State where Care Stat recorded [73(263%)] positive malaria cases and SD Bioline recorded [70(25.2%)].

Comparison between Microscopy and the RDTs used in the study

Comparison between microscopy and the RDTs in Tables 5 show the values of the true and false positive, and negative results which were used in the calculations of sensitivity, specificity and predictive values.

Table 1: prevalence of malaria parasite in blood sample of patients attending Nasarawa state university hospital, keffi and General Hospital Kawo Kaduna.

Location	Gender	No. examined	No. Positive	No. Negative	Percentage Prevalence (%)	P-value
Nasarawa (N=377)	**Female**	220	100	120	45.5	*P* = 0.016
	Male	157	52	105	33.1	
$\chi^2 =$ 5.7918 df=1						
TOTAL		377	152	225	40.3	
Kaduna N=278						*P* =0.047
	Female	130	70	60	53.8	
	Male	148	97	51	65.5	
TOTAL		278	167	111	39.9	

KEY: % = PERCENTAGE, N = NUMBER $\chi^2 = 3.9461$ df=1

Table 2: Prevalence of Malaria parasite infection by Microscopy and RDTs among patients attending Nasarawa State University hospital, Keffi and General Hospital Kawo, Kaduna

Location	Gender	No. examined	No. positive RDT	No. positive Microscopy	Prevalence rate RDT (%)	Prevalence rate Microscopy (%)
Kaduna	**Female**	130	30	40	10.8	14.4
	Male	148	43	54	15.5	19.4
TOTAL		278	73	94	26.3	33.8
Nasarawa						
	Female	157	44	56	11.7	14.9
	Male	220	29	23	7.7	6.1
TOTAL		377	73	79	19.4	21.0

KEY: RDTS = RAPID DIAGNOSTIC TEST, N= NUMBER

Diagnostic performance of rapid diagnostic tests used

Comparing the diagnostic performance of the two RDTs Carestat and SD Bioline used in Keffi, Nasarawa State gave a sensitivity and specificity of 45.1% and 96.6% respectively (Table 5). The Positive Predictive Value (PPV) and Negative Predictive Value (NPV) were found to be 89.0% and 74.0% respectively. The comparisons of the sensitivity, specificity, positive predictive value and negative predictive value of the rapid diagnostics test kits used in Kaduna State health care Centre, showed that Carestat had a sensitivity and specificity of 91% and 98% respectively, positive predictive value of 95% and negative predictive value of 98%. SD Bioline on the other hand sensitivity of 96%, specificity of 100% positive predictive value and negative predictive value of 99%.

Table 3: Relative malaria percentage positive and negative values by RDT kits and microscopy in Nasarawa State University hospital, Keffi and General hospital, Kawo, Kaduna.

OUTCOMES	TECHNIQUES		
Nasarawa	**MICROSCOPY (%)**	**CARESTAT® N (%)**	**SD BIOLINE® N (%)**
Positive	79(21.0)	73(19.4)	73(19.4)
Negative	298 (79.0)	304(80.6)	304(80.6)
Total	377(100)	377(100)	377(100)
Kaduna			
Positive	94(33.8)	73(26.3)	70(25.2)
Negative	184(66.2)	205(73.7)	208(74.8)
Total	278(100)	278(100)	278(100)

KEY: RTD = RAPID DIAGNOSTIC TEST, %= PERCENTAGE

Table 4: Comparison between Microscopy and RDTs used at the Nasarawa State University hospital Keffi

RDT	**MICROSCOPY**		
	POSITIVE (%)	**NEGATIVE (%)**	**TOTAL (%)**
POSITIVE (%)	65 (45.1) TP	8(3.4) FP	73(19.4)
NEGATIVE (%)	79 (54.9) FN	225(96.6) TN	304(80.6)
TOTAL (%)	144 (100)	233(100)	377(100)

KEY: TP = TRUE POSITIVE, FP= FALSE POSITIVE, FN = FALSE NEGATIVE, TN =TRUE NEGATIVE

Overall, a stark contrast in performance was observed between the two healthcare centers. In Nasarawa, both RDTs showed identical, low sensitivity (44.3%) but high specificity (97.3%). In Kaduna, both tests performed excellently, with SD Bioline showing superior sensitivity (95.8% vs. 90.8%) and specificity (99.5% vs. 98.0%) compared to CareStart.

DISCUSSION

According to the 2021 World Malaria Report, Nigeria had the highest number of global malaria cases (27%) and global malaria deaths (32%) in 2020. Nigeria also accounted for an estimated 55.2% of malaria cases in West Africa in the same year.[4] These figures underscore the significant malaria burden in the country and the need for accurate diagnosis and effective control measures.

In this study, malaria prevalence was 39.9% in General hospital Kawo-Kaduna and 40.3% in Nasarawa State University Health Centre. These values confirm that malaria remains highly endemic in these northern states despite the presence of intervention programs aimed at halting transmission. The prevalence recorded in this study is higher than the 21.1% reported in Ethiopia[21] and higher than the microscopy data from the 2018 Nigeria Demographic and Health Survey (NDHS), which reported prevalence rates ranging from 16% in the Southeast zones to

34% in the Northwest zone.[21] However, our findings were lower than those reported in Kano (64.9%) and Ondo State (82.7%)[22-23], and comparable to studies conducted in Rivers and Benue States.[24-25] A lower prevalence has previously been reported for Nasarawa State.[26] These variations may be due to ecological differences, such as poor drainage and stagnant water bodies that favor mosquito breeding, in addition to antimalarial drug resistance, which contributes to sustained transmission in endemic regions.

A statistically significant difference in malaria prevalence between genders was observed ($P < 0.05$). In Kaduna, malaria parasitaemia was higher among males, whereas in Nasarawa, females had slightly higher prevalence. The biological reasons for these patterns are not completely clear. Some studies suggest sex-specific differences in naturally acquired immunity, with males being less immune to controlling parasite densities, thereby leading to higher parasite prevalence.[27]Conversely, Gill and Warell[28] reported no empirical evidence supporting a gender predilection to malaria infection. Similar gender-related trends have been observed in Uganda, where malaria positivity was slightly higher among males using either RDT or microscopy.[29] These findings suggest that both biological and socio-cultural factors may play a role in exposure and infection rates.

Microscopy detected the highest number of positive cases, with 33.8% positivity in Kaduna and 21% in Nasarawa, further confirming its position as the gold standard for malaria diagnosis. This finding aligns with previous reports across Nigeria (Rivers, Benue and Ibadan).[2, 25, 30]

When comparing RDTs to microscopy, distinct differences were observed between the two study locations. In Kaduna, Carestat and SD Bioline RDTs demonstrated high sensitivity (91% and 96%), specificity (98% and 100%), positive predictive value (95% and 99%), and negative predictive value (98% and 99%), indicating excellent performance that aligns with WHO standards and other studies conducted in FCT, Keffi, and Rivers State.[31-32] Conversely, in Nasarawa, both RDTs showed much lower sensitivity (45.1%), while specificity (96.6%) and PPV (89%) remained high.

High specificity indicates that false positives were rare, which is important for avoiding unnecessary treatment. However, the low sensitivity implies that more than half of true malaria cases were missed, posing a serious public health risk. This mismatch between sensitivity and specificity shows that while a positive test result is reliable, a negative result cannot be trusted without further confirmation.

The most plausible explanation for the poor sensitivity in Nasarawa is degradation of RDT kits due to improper storage conditions; the erratic power supply as well as the hot weather might have affected the storage condition of these kits. RDTs are highly sensitive to temperature and humidity, and poor supply chain management can compromise their integrity before use.[35] This is a common problem in tropical climates and highlights the importance of maintaining a continuous cool chain during transportation and storage.

Other factors, such as batch-to-batch variations or differences in parasite density between populations, may also contribute, though they are less likely to cause the dramatic decline in sensitivity observed in this study. The similarity of results between Carestat and SD Bioline within the same location suggests that environmental and logistical factors are more likely responsible than intrinsic test quality.

The strong performance of RDTs in Kaduna mirrors findings from studies conducted in Ibadan-Oyo State and Uganda where two RDT brands also performed well alongside microscopy.[29-30]

Similarly, reports from FCT and Rivers State demonstrated comparable accuracy and reliability.[31-33]

Table 5: Comparison between Microscopy and the RDTs used in General Hospital Kawo, Kadu

	MICROSCOPY		
RDT Carestat®	**POSITIVE (%)**	**NEGATIVE (%)**	**TOTAL**
Positive (%)	69(91) TP	4(1.9) FP	73(26.3)
Negative (%)	7(9) FN	198(98.1) TN	205(73.7)
Total	76(100)	202(100)	278(100)
SD BIOLIN® N RDT			
Positive (%)	69(95.8) TP	1(0.5) FP	70(25.2)
Negative (%)	3(4.2) FN	205(99.5) TN	208(74.8)
Total	72(100)	206(100)	278(100)

KEY: TP = TRUE POSITIVE, FP= FALSE POSITIVE, FN = FALSE NEGATIVE, TN =TRUE NEGATIVE

In contrast, a study from Jos using Carestat reported sensitivity and specificity of 78.4% and 97.6%, respectively, with a PPV of 93.3% and NPV of 80.1%, which are lower than the values observed in Kaduna.[34] These variations across different locations within Nigeria further emphasize the role of local environmental factors, supply chain quality, and test handling practices in determining diagnostic performance.

The low sensitivity recorded in Nasarawa is particularly concerning. Missed diagnoses lead to untreated malaria cases, prolonged illness, and sustained transmission. In high-burden settings like Nasarawa, negative RDT results should be confirmed by microscopy, especially for patients with strong clinical symptoms. This dual-approach strategy will help minimize missed diagnoses and reduce malaria-related morbidity and mortality.

This study reaffirms that while RDTs are useful in emergency situations and areas with erratic power supply, microscopy remains indispensable in facilities where quality assurance can be maintained.

Table 6: Diagnostic performance of the RDTs used in Table 6: Diagnostic performance of the RDTs used in Nasarawa State University hospital and General Hospital Kawo, Kaduna

PERFORMANCE	**RDTS**	
Nasarawa		
	CARESTAT (%)	SD BIOLINE (%)
Sensitivity TP/TP+FN *100	90.7	95.8
Specificity TN/FP+TN*100	98.0	99.5
Positive Predictive Value TP/TP+FP*100	94.5	98.6
Negative Predictive value TN/TN+FN*100	97.5	98.6
Kaduna		
	CARESTAT (%)	SD BIOLINE (%)
Sensitivity TP/TP+FN *100	45.1	45.1
Specificity TN/FP+TN*100	96.6	96.6
Positive Predictive Value TP/TP+FP*100	89.0	89.0
Negative Predictive value TN/TN+FN*100	74.0	74.0

KEY: RDTS = RAPID DIAGNOSTIC TEST, TP = TRUE POSITIVE, FP= FALSE POSITIVE, FN = FALSE NEGATIVE, TN =TRUE NEGATIVE

CONCLUSION

This study confirms the high prevalence of malaria in the two health centers of Kaduna and Nasarawa States, emphasizing the continued public health threat in northern Nigeria. While RDTs demonstrated excellent performance in Kaduna, their significantly reduced sensitivity in Nasarawa highlights a major weakness in the diagnostic pipeline, most likely due to suboptimal storage and handling conditions.

Although RDTs are critical for rapid malaria screening, microscopy remains the most reliable tool for accurate detection. Strengthening both diagnostic approaches will improve treatment outcomes, prevent missed diagnoses, and support malaria elimination efforts in Nigeria

In order to enhance diagnostic reliability and bolster malaria control efforts, a multi-faceted strategy is imperative. This must begin with strengthening supply chain logistics to ensure a continuous cool chain for RDTs from manufacturer to point-of-use, preserving their efficacy. Complementing this, the implementation of rigorous quality assurance protocols, including regular lot testing and monitoring, is essential to validate kit performance. Furthermore, continuous training for healthcare workers on proper RDT handling, interpretation, and an understanding of their limitations is crucial. Given the risk of false-negative results, a policy of mandatory microscopy confirmation for negative RDTs in high-transmission areas or for symptomatic patients is a critical safety measure. While RDTs are indispensable for rapid screening, microscopy remains the most reliable diagnostic tool. Ultimately, strengthening both diagnostic approaches in tandem will significantly improve treatment outcomes, prevent missed diagnoses, and accelerate progress toward malaria elimination.

REFERENCES

1. Yan SLR., Wakasuqui, F & Wrenger, C. Point-of-care tests for malaria: Speeding up the diagnostics at the bedside and challenges in malaria cases detection. Diagnostic Microbiology and Infectious Disease 2020; 98(3),115122. https://doi.org/10.1016/j.diagmicrobio.2020.115122
2. Ken-Ezihuo, U. S., Itatat, S. S., & Ebirien-Agana, S. B. Comparative study of two different rapid diagnostic tests with microscopy method for malaria parasite detection. Journal of Advances in Medicine and Medical Research 2019; 30(9), 1-8. https://doi.org/10.9734/jammr/2019/v30i930230
3. World Health Organization. (2019). World malaria report 2019. https://apps.who.int/iris/handle/10665/330011
4. World Health Organization. (2021). World malaria report 2021. https://www.who.int/teams/global-malaria-programme/reports/world-malaria- report-2021
5. Bhandari, P. L., Raghuveer, C. V., Rajeev, A., & Bhandari, P. D. Comparative study of peripheral blood smear, quantitative buffy coat and modified centrifuged blood smear in malaria diagnosis. Indian Journal of Pathology and Microbiology, 2018 51(1),108–112. https://doi.org/10.4103/0377-4929.40421
6. World Health Organization. (2010). Guidelines for the treatment of malaria (2nd ed.). https://www.who.int/publications/i/item/9789241547925
7. Tilley, L., Dixon, M. W., & Kirk, K. (2011). The Plasmodium falciparum-infected red blood cell. International Journal of Biochemistry and Cell Biology, 43(6), 839–842. https://doi.org/10.1016/j.biocel.2011.03.011
8. Endeshaw, T., Gebre, T., Ngondi, J., Graves, P. M., Shargie, E. B., Ejigsemahu, Y., Ayele, B., Yohannes, G., Tilahun, A., Mesfin, A., Teferi, T., Zerihun, M., Genet, A., Mosher, A. W., Emerson, P. M.,

& Richards, F. O. (2008). Evaluation of light microscopy and rapid diagnostic test for the detection of malaria under operational field conditions: a household survey in Ethiopia. Malaria Journal, 7, 118. https://doi.org/10.1186/1475-2875-7-118

9. Oboh, M. A., Fatou, F., Oluwagbemisola, E. A., Simon, C., Anthony, U. A., Ekon, O., & Ntadom, G. (2022). Multiple Plasmodium falciparum drug resistance polymorphisms identified in a pregnant woman with severe malaria and a concomitant spontaneous abortion in Cross River, Nigeria, West Africa. Malaria Journal, 21(1), 160. https://doi.org/10.1186/s12936-022-04184-9
10. Berhane A, Russom M, Bahta I, Hagos F, Ghirmai M, Udhayakumar, V & Lucchi NW. *pfhrp2/3* gene deletions in Plasmodium falciparum parasites from Eritrea and other African countries. Malaria Journal 2021; 20(1), 1–10. https://doi.org/10.1186/s12936-021-03717-5
11. Landier, J., Parker, D. M., Thu, A. M., Carrara, V. I., Lwin, K. M., Bonnington, C. A., ... & Smithuis, F. (2018). The role of ultrasensitive rapid diagnostic tests in malaria control and elimination: A critical review. Malaria Journal, 17(1), 1–11. https://doi.org/10.1186/s12936-018-2471-3
12. Odugbemi, B. A., Adewunmi, O. A., Akinwale, O. P., Okoh, F., Olukosi, Y., & Fatunmbi, B. (2019). Diagnostic accuracy of malaria rapid diagnostic tests in Nigeria: Findings from the 2015 Nigeria Malaria Indicator Survey. Malaria Journal, 18(1), 1–10. https://doi.org/10.1186/s12936-019-2916-5
13. World Health Organization. (2023). World malaria report 2023. https://www.who.int/teams/global-malaria-programme/reports/world-malaria- report-2023
14. World Health Organization. (2023). Diagnostic testing for malaria. https://www.who.int/activities/diagnostic-testing-for-malaria
15. Akwa, V. L., Binbol, N. L., Samaila, K. L., & Marcus, N. D. (2007). Geographical perspective of Nasarawa State. Onaivi Printing and Publishing Company Limited.
16. Geodatos. (2023). Geographic coordinates of Kaduna, Nigeria. https://www.geodatos.net/en/coordinates/nigeria/kaduna
17. Naing, L., Winn, T., & Rusli, B. N. (2006). Practical issues in calculating the sample size for prevalence studies. Archives of Orofacial Sciences, 1, 9–14.
18. Jemimah, Y., Oti, V., Amuta, E., Philip, A., & Anizoba, L. (2019). Plasmodium falciparum infection among febrile patients attending a tertiary healthcare facility in Central Nigeria: Prevalence, hematologic and socio-demographic factors. International Journal of Tropical Disease, 2(2), 1–6.
19. Bajoga, U. A., Balarabe, H. S., Olufemi, A. A., Dalhat, M. M., Sule, I. B., Ibrahim, M., ... & Nguku, P. (2019). The Pan African medical journal, 32(Suppl 1), 8. https://doi.org/10.11604/pamj.supp.2019.32.1.13735
20. Cheesbrough, M. (2005). District laboratory practice in tropical countries, Part 2. Cambridge University Press.
21. President's Malaria Initiative. (2022). Nigeria malaria operational plan FY 2022. https://d1u4sg1s9ptc4z.cloudfront.net/uploads/2022/03/FY-2022-

Nigeria- Malaria-Operational-Plan.pdf

22. Oladele, O. V., Onuoha, S. C., Hamafyelto, H. S., Omisope, O., Fauziyya, A., Akindigh, M., & Sabiu, L. (2018). Prevalence of malaria infection among patients attending Murtala Muhammed specialist hospital Kano, Nigeria. African Journal of Clinical and Experimental Microbiology, 19(3), 215–220. https://doi.org/10.4314/ajcem.v19i3.6
23. Omoya, F., & Ajayi, K. (2020). Prevalence of malaria among febrile patients attending government hospitals in Ondo State, South-West Nigeria. American Journal of Epidemiology and Public Health, 4(4), 17–24.
24. Egbom, S. E., Nduka, F. O., & Nzeako, S. O. (2022). Point prevalence mapping of malaria infection in Rivers State, Nigeria. Tanzania Journal of Health Research, 23(4). https://doi.org/10.4314/thrb.v23i4.x
25. Okita, F. O., & Amuta, E. U. (2017). Comparative use of RDT and thick film microscopy in the diagnosis of malaria in sub-urban settlements in Makurdi, Nigeria. International Journal of Infectious Diseases and Therapy, 2(2), 25–34. https://doi.org/10.11648/j.ijidt.20170202.11
26. Bello, B. O., & Ishaleku, D. (2018). Prevalence of malaria infection among people living with HIV/AIDS at Federal Medical Center Keffi (Nasarawa State), Nigeria. Journal of Advances in Medicine and Medical Research, 11(4), 1–6. https://doi.org/10.9734/JAMMR/2018/43154
27. Rodriguez-Barraquer, I., Arinaitwe, E., Jagannathan, P., Kamya, M. R., Rosenthal, P. J., Rek, J.,... & Dorsey, G. (2018). Quantification of anti-parasite and anti-disease immunity to malaria as a function of age and exposure. eLife, 7, e35832. https://doi.org/10.7554/eLife.35832
28. Gilles, H. M., & Warrell, D. A. (1993). Bruce-Chwatt's essential malariology (3rd ed.). Heinemann Medical Books.
29. Okiring, J., Epstein, A., Namuganga, J. F., Kamya, E. V., Nabende, I., Nassali, M., ... & Kamya, M. R. (2022). Gender difference in the incidence of malaria diagnosed at public health facilities in Uganda. Malaria Journal, 21(1), 22. https://doi.org/10.1186/s12936-022-04046-4
30. Ilesanmi, R. I., Olalubi, O. A., Adetunde, O. T., Ilesanmi, A. O., Effedua, H., & Amoo, A. O. (2017). Comparative assessment of malaria rapid diagnostic test (RDT) in Ibadan, Nigeria. Malaria World Journal, 8(17), 1–7.
31. Mac, P. A., Asheadzi, H. F., Gideon, A., Thaker, P., & Airiohuodion, P. (2019). Prevalence of Plasmodium falciparum among Nigerians in Abuja and Central States: A comparative analysis of sensitivity and specificity using rapid diagnostic test and microscopy as tools in management of malaria. International Journal of Tropical Disease, 1(1), 014. https://doi.org/10.23937/ijtd-2017/1710014
32. Micheal W & Florence N. Validity of presumptive diagnosis of malaria among outpatients in a tertiary healthcare facility in rivers state, Nigeria. British Journal of Medical and Health Research 2016; 3(5): 52–7.
33. Olasehinde, G. I., Oyeka, U. C., Oniha, M. I., Onile-Ere, O. A., Ayepola, O. O.,

Ajayi, A. A., & Ojurongbe, O. Data set on Rapid Diagnostic Tests (RDTs) and microscopy for diagnosing plasmodium falciparum and plasmodium vivax. Data in Brief 2018; 20: 503–9. https://doi.org/10.1016/j.dib.2018.08.032

34. Sheyin, Z., & Bigwan, I. E. Comparison of care start HRP2 rapid malaria test with light microscopy for guiding patients' treatment of fever in Nigerian endemic areas. Journal of Medicine and Medical Sciences 2013; 4(9), 353–6.

35. Jorgensen, P., Chanthap, L., Rebueno, A., Tsuyuoka, R., & Bell, D. Malaria rapid diagnostic tests in tropical climates: The need for a cool chain. The American Journal of Tropical Medicine and Hygiene 2006; 74(5), 750–4. https://doi.org/10.4269/ajtmh.2006.74.750